
AMPERE DEVELOPMENT

Power Testing & Load Bank Solutions

Why Do I Need a Load Bank to Test My Generators?

Why Starting the Engine Is Not the Same as Proving the System

A generator that starts, runs quietly, and shows good voltage at no load has not been tested. It has only been idled. When utility power fails, that same machine must suddenly produce rated kilowatts, hold frequency, stay cool, and carry real demand. A load bank is the only practical way to apply that demand on purpose, under controlled conditions, with results you can document.

If you only run the generator unloaded, you have not proven it will work when you need it.

The Problem with No-Load and Light-Load Running

Many facilities “exercise” generators monthly by starting them and letting them run for a short time. That is useful for lubrication and a basic start check. It is not a load test. Light-load running can actually create problems of its own:

- Unburned fuel and carbon collect in the exhaust (wet stacking)
- Cylinder glaze, increased oil consumption, and poor combustion efficiency
- Cooling, fuel, and exhaust systems never reach the temperatures they see in an emergency
- Voltage regulators, governors, and transfer equipment are never stressed
- Hidden faults stay hidden until the next real outage

Why Building Load Usually Is Not Enough

Using the building as the load sounds convenient, but it is rarely adequate or safe as a complete test:

- Typical building demand is often far below nameplate kW, especially nights, weekends, or in lightly occupied facilities
- You cannot control the load profile or hold a required percentage for a required duration
- You risk disrupting operations if the test goes poorly
- Codes such as NFPA 110 require specific load levels and times that building load often cannot meet
- You still need a way to add load if the building cannot reach 30%, 50%, 75%, or 100%

Eight Reasons You Need a Load Bank

1. You Must Prove Rated Capacity

Only a load bank (or an equivalent, fully controllable load) can take the generator to a known percentage of nameplate kW and hold it there. That is the only way to confirm the engine and alternator can actually deliver what the nameplate claims.

2. Wet Stacking Will Damage a Lightly Loaded Diesel

Diesels designed for high load that spend most of their life at idle or 10–20% load accumulate unburned fuel in the exhaust. Staged or full-load testing burns off those deposits, restores exhaust temperature, and helps prevent turbo, injector, and stack damage.

3. Codes and Inspectors Require It

NFPA 110 and many AHJs require periodic loaded exercise and annual tests at defined percentages and durations. Hospitals face additional Joint Commission expectations. A load bank is how you reach those numbers when building load cannot.

4. Cooling, Fuel, and Exhaust Only Work Under Load

Thermostats, radiators, fuel pumps, and exhaust systems are designed around real heat and fuel flow. No-load running never exercises them the way an outage will. Load-bank testing finds plugged radiators, weak pumps, and undersized exhaust before an emergency does.

5. Electrical Stability Matters as Much as Starting

You need stable voltage and frequency as load is added and removed. Load banks let you apply steps, watch AVR and governor response, and catch hunting, sag, or imbalance that a no-load start will never show.

6. Transfer Switches and Paralleling Gear Need Load Too

Automatic transfer, paralleling, and load-shed schemes must be proven under demand—not just by watching lights come on. A load bank lets you test those sequences without gambling on live facility load.

7. New or Repaired Sets Need Acceptance Testing

After installation, major repair, or relocation, a load bank is the standard way to accept the generator on site. Factory tests are not a substitute for as-installed performance at your altitude, temperature, and fuel system.

8. Documentation Protects You

A recorded load test is evidence for insurance, due diligence, warranties, and compliance. “It started last month” is not evidence. Voltage, kW, frequency, and temperature logs are.

Typical Generator Load-Test Targets

Purpose	Typical Approach
Monthly exercise (NFPA 110)	Minimum ~30% nameplate kW, or manufacturer exhaust-temp method, ≥30 minutes
Annual load test (current NFPA 110 reference)	50% for 30 minutes, then 75% for 60 minutes
Commissioning / acceptance	Staged 25% → 50% → 75% → 100%, with a heat-soak at high load
After repair or overhaul	Staged load to confirm repaired systems under stress
Wet-stacking cleanup	Sustained higher load until exhaust and combustion stabilize

Always confirm the exact protocol required by your manufacturer, AHJ, and current edition of applicable standards.

The Bottom Line

You need a load bank because an emergency generator’s job is not to start. Its job is to carry load. A load bank is how you apply that load safely, measure it accurately, meet code, protect the engine, and leave with a record that the set will perform. Renting from Ampere Development gives you the right capacity and configuration without buying equipment you only need a few times a year.

Need to load-test a generator?

Tell us the kW, voltage, and test goal. We will size the rental and help you plan the sequence.

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